

Portfolio Construction: From Fragile Optimisation to Robust Allocation

Abstract: Building on the findings that classical sample mean-variance optimisation is highly sensitive to estimation error and often fails to outperform the naïve $1/n$ benchmark out-of-sample, this instalment examines whether robust variants can restore some of its practical relevance. Specifically, it focusses on three core approaches: covariance and mean shrinkage, Bayesian priors and explicit weight constraints. The results highlight that portfolio performance is not driven by optimisation itself, but by how effectively estimation error is controlled.

Michaud (1989) famously demonstrated that the sample-based mean-variance optimiser does not merely underperform: it is **error-maximising** rather than Sharpe-maximising. To see why, recall the standard mean-variance problem for a given target return μ_p :

$$\min_w w^\top \hat{\Sigma} w \quad \text{subject to} \quad w^\top \hat{\mu} = \mu_p, \quad \mathbf{1}^\top w = 1,$$

where $\hat{\mu}$ and $\hat{\Sigma}$ are sample estimates. The well-known solution is:

$$w^* = \frac{\mu_p}{a} \hat{\Sigma}^{-1} \hat{\mu} + (\text{terms for the unit constraint}),$$

with $a = \hat{\mu}^\top \hat{\Sigma}^{-1} \hat{\mu}$. Estimation errors in $\hat{\mu}$ are multiplied by $\hat{\Sigma}^{-1}$. Because the inverse covariance matrix magnifies small perturbations, especially when assets are numerous and correlated, the optimiser assigns extreme bets to assets with even modestly overestimated returns. In practice, this renders the classical Markowitz portfolio strikingly poor out-of-sample.

The first successful remedy came from shrinking expected returns. In a 1991 paper, Philippe Jorion showed that pulling historical sample means towards a common grand mean, the James-Stein shrinkage estimator, consistently improved out-of-sample performance. Formally, the shrunk return vector is:

$$\tilde{\mu} = (1 - \lambda) \hat{\mu} + \lambda \mu_0 \mathbf{1},$$

where μ_0 is the grand mean and λ is a shrinkage intensity. This insight directly inspired the Black-Litterman model (Black and Litterman, 1992). Instead of relying on raw sample means, Black and Litterman start from a market equilibrium prior μ^{eq} and blend it with investor views:

$$\tilde{\mu} = \left[(\tau \Sigma)^{-1} + P^\top \Omega^{-1} P \right]^{-1} \left[(\tau \Sigma)^{-1} \mu^{\text{eq}} + P^\top \Omega^{-1} Q \right],$$

where P picks the assets involved in views, Q contains the view returns, and Ω measures view uncertainty. The resulting expected returns are far more stable and less prone to optimisation extremes.

Around the same time, researchers tackled the covariance matrix. Ledoit and Wolf (2003, 2004) introduced linear shrinkage of the sample covariance matrix toward a structured target F , such as the constant-correlation model:

$$\tilde{\Sigma} = \delta F + (1 - \delta) \hat{\Sigma}, \quad \delta \in [0, 1].$$

While their estimator can be interpreted as having a Bayesian prior, Ledoit and Wolf emphasised that shrinkage works distribution-freely: it systematically reduces the most harmful estimation errors without imposing any parametric assumption on returns. This made covariance shrinkage both robust and broadly applicable to many other fields using covariance estimation in their process.

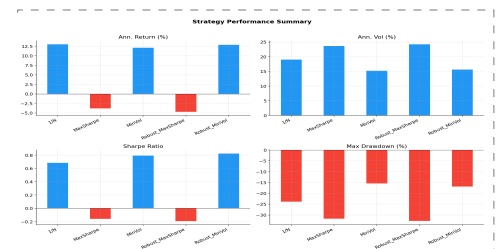
A different, more direct route emerged from Jagannathan and Ma (2003). They proved that imposing simple weight constraints, such as a no-short-sale constraint $w_i \geq 0$ or a norm-based bound $\|w\|_2^2 \leq c$, acts as an implicit form of shrinkage. By restricting the feasible set $\mathcal{W}$, constraints reduce the effective number of free parameters and stabilise the optimisation without ever modifying $\hat{\mu}$ or $\hat{\Sigma}$. The constrained problem's solution becomes equivalent to a regularised estimator. This finding neatly unified earlier approaches: constraints, shrinkage on means and shrinkage on covariances all impose structure on an otherwise ill-posed optimisation problem. Despite these advances, the $1/n$ benchmark still remained a surprisingly difficult hurdle as DeMiguel, Garlappi and Uppal showed in 2007.

(The next instalment introduces Hierarchical Risk Parity, a framework that avoids direct inversion of the covariance matrix and instead allocates capital based on the hierarchical structure of asset covariances. Therefore, bypassing error amplification and offering a different approach to robustness.)

Charts

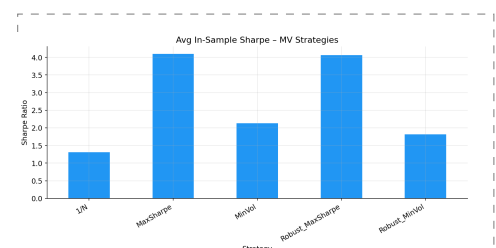
Out-of-sample Performance Summary

Performance dashboard showing improvements and trade-offs



Classical vs Robust In-sample Sharpe Ratio

Example of classical approach overestimating in sample statistics



Source: prometheuscaptial.nl

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